DO NOW — RECALL

Lesson 091 · Fetch–Decode–Execute: A Register Trace

Name: _____ Class: _____ Date: _____

1. Which register holds the current instruction?

2. What is the role of MAR?

3. Where is the next instruction address kept?

4. A full adder has A=1,B=0 and carry-in=1. State Sum and carry-out.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

CIR.

2. Model answer

Stores address being accessed.

3. Model answer

PC.

4. Model answer

$1+0+1=10_2$, so Sum0, carry-out1.

By the end, you can…

• Fetch copies the address from PC to MAR; memory read transfers the instruction through MDR into CIR. In a sequential fetch, PC advances.

• The Control Unit decodes the current instruction held in CIR and produces the necessary control signals.

• Execution of LDA 80 reads the data held at memory address 80 into ACC; the next instruction is not executed until its own fetch.

Vocabulary — know these terms

fetch — Retrieve next instruction.

decode — CU interprets instruction.

execute — Carry out operation.

CIR — Current Instruction Register.

ACC — Accumulator.

NEW LEARNING

Read, annotate and explain

Observe

Fetch copies the address from PC to MAR; memory read transfers the instruction through MDR into CIR. In a sequential fetch, PC advances.

Explain

The Control Unit decodes the current instruction held in CIR and produces the necessary control signals.

Evaluate

Execution of LDA 80 reads the data held at memory address 80 into ACC; the next instruction is not executed until its own fetch.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

PC=30, memory[30]=ADD 70, memory[70]=5, ACC=4. Trace after fetch and execute.

Explain the mechanism

Fetch: MAR30, CIR ADD70, PC31. Execution reads memory[70]=5 and ALU adds to ACC4, giving ACC9.

The PC increments in fetch while the ALU operates on ACC and fetched operand.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Trace the fetch of instruction at PC20 into MAR,MDR,CIR; give PC after sequential increment. [6]

Question 2

Explain difference between decode and execute for LDA 80 when memory[80]=13. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

An answer states 'MDR stores address 80 and MAR stores the fetched 13'. Correct the error with a precise memory-read explanation. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

During a sequential fetch, MDR stores the instruction transferred from memory rather than its address.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.